

Make UploadFile check for future rollover

#2962

New issue

Merged

Kludex merged 9 commits into `Kludex:master` from `HonakerM:fix_upload_file_hang` on Jul 21, 2025

Conversation11

Commits9

Checks7

Files changed2

+82-6



HonakerM commented on Jul 10, 2025

Contributor

• edited

Summary

This PR ensures that during multi-part form parsing any slow disk IO happens in a background. This specifically fixes a case where a file would rollover during a `write()` call as talked about in [this discussion](#).

Checklist

- ☒ I understand that this PR may be closed in case there was no previous discussion. (This doesn't apply to typos!)
- ☒ I've added a test for each change that was introduced, and I tried as much as possible to make a single atomic change.
- ☐ I've updated the documentation accordingly.

Reviewers

- Kludex ✓
- defnull ●

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

3 participants



HonakerM added 4 commits 6 months ago

- [Make UploadFile check for future rollover](#) [0aff77a](#)
- [Fix Linting/MyPy Errors](#) [4fcf16e](#)
- [Add test for large multi-part file](#) [6fc62ab](#)
- [Fix Coverage Lines](#) [e3cffca](#)



 **HonakerM** marked this pull request as ready for review
6 months ago



 **Slightly improve efficiency of Size Check**  [f14ee57](#)



Kludex reviewed [on Jul 11, 2025](#) [View reviewed changes](#)

tests/test_formparsers.py **Outdated**

Comment on lines 307 to 314

```
307 + def test_multipart_request_large_file():
308 +     data = BytesIO(b" " * MultiPartFormParser.DEFAULT_MAX_SIZE)
309 +     client = test_client_factory(application)
310 +     response = client.post(
311 +         "/",
312 +         files=[("test_large", data)],
313 +     )
314 +     assert response.status_code == 200
```



Kludex [on Jul 11, 2025](#)

Owner

This test passes on master. We want some condition that proves this PR actually solves the issue.



defnull [on Jul 11, 2025](#)

This type of issue is hard to verify without profiling or adding breakpoints and checking if a write operation that caused a rollover happened on the main thread. I do not know if it is feasible or even possible to provide a unit test that fails if the blocking rollover happens on the wrong thread, that would require assert statements and additional logic in the actual code.



Kludex [on Jul 11, 2025](#)

Owner

Then we can remove the test, can't we?



HonakerM

Contributor

Author

[on Jul 11, 2025](#) • edited ▼

[@Kludex](#) yeah in it's current state it's not doing a whole lot. Let met try editing the test to use a monkey patched SpooledTemporaryFile to ensure the IO thread is separate from the event thread



defnull [on Jul 11, 2025](#)

It does not ensure that there is no blocking call on the event loop, but it does cover the new code paths that would otherwise not be covered.



HonakerM

Contributor

Author

on Jul 11, 2025

@Kludex Just pushed the new test! I ended up creating a new class/app to track where rollover was happening. I'm still new to starlette development so please let me know if you have any code-quality suggestions.



HonakerM added 2 commits [6 months ago](#)



Update test to actually test the bug ✗ [753e790](#)



Fix test mypy issues ✓ [297c09e](#)



HonakerM [force-pushed](#) the [fix_upload_file_hang](#) branch from [4a27556](#) to [297c09e](#) [6 months ago](#) [Compare](#)



defnull commented [on Jul 11, 2025](#)

Not for this PR, but at this point I would just move the rollover logic into `UploadFile` and not use `SpooledTemporaryFile` at all. That would also save some expensive `getattr()` calls in the hot path. Maybe an idea for a future PR.



HonakerM commented

Contributor

Author

[on Jul 11, 2025](#)

@defnull @Kludex mentioned this elsewhere but [PR2925](#) is probably the long term solution here. It moves all this logic to `anyio` and gets it out of the hot path



Merge branch 'master' into [fix_upload_file_hang](#) Verified ✓ [f1fee10](#)



HonakerM requested review from **Kludex** and **defnull** [6 months ago](#)



HonakerM commented

Contributor

Author

on Jul 16, 2025

@Kludex sorry for the ping/re-review request but I'm wondering if we could get this merged soon? It should be ready to go. Please let me know if you have any more comments/suggestions/requirements



Kludex commented on Jul 17, 2025

Owner

I'll release it today.



modify opinionated changes

✓ 50ae503



Kludex approved these changes
on Jul 21, 2025

[View reviewed changes](#)



Kludex merged commit **9f7ec2e** into
Kludex:master on Jul 21, 2025
7 checks passed

[View details](#)



HonakerM commented

Contributor

Author

on Jul 21, 2025

@Kludex since you published the security bulletin I'm no longer able to comment so I'm going to follow-up here:

1. wai25 is one of my co-workers who brought the issue to my attention and helped me narrow down the problem to starlette.
2. Could we request a low CVE for this issue? It seems the other advisories had them created



github-actions bot mentioned this pull request
on Jul 28, 2025

Dependency audit detected
vulnerabilities adrianwedd/agenti-
research-engine#497

✓ Closed

 **github-actions** (bot) mentioned this pull request
on Aug 4, 2025

Dependency audit detected
vulnerabilities [adrianwedd/agenti-](#)
[research-engine#498](#)

✓ Closed



 **depthfirst-dev** (bot) mentioned this pull request
on Oct 24, 2025

[depthfirst-5909] Upgrade multiple
packages in uv.lock [seshanktest/tts#2](#)

🔗 Open



This was referenced on Dec 15, 2025

📄 **Bump versions of multiple**
dependencies to address
vulnerabilities [muditsinha01/multi-lang-](#)
[minimal#19](#)

🔗 Open

📄 **Bump versions of multiple**
dependencies to address
vulnerabilities [muditsinha01/multi-lang-](#)
[minimal#20](#)

🔗 Open

📄 **Bump versions of multiple**
dependencies to address
vulnerabilities [muditsinha01/multi-lang-](#)
[minimal#21](#)

🔗 Open

📄 **Bump versions of multiple**
dependencies to address
vulnerabilities [muditsinha01/multi-lang-](#)
[minimal#22](#)

🔗 Open

📄 **Bump versions of multiple**
dependencies to address
vulnerabilities [muditsinha01/multi-lang-](#)
[minimal#23](#)

🔗 Open

📄 **Bump versions of multiple**
dependencies to address
vulnerabilities [muditsinha01/multi-lang-](#)
[minimal#24](#)

🔗 Open



This was referenced on Dec 15, 2025

📄 **Bump versions of multiple**
dependencies to address
vulnerabilities [muditsinha01/multi-lang-](#)
[minimal#25](#)

🔗 Open

[Bump versions of multiple dependencies to address vulnerabilities](#) [muditsinha01/multi-lang-minimal#26](#)

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[Bump versions of multiple dependencies to address vulnerabilities](#) [muditsinha01/multi-lang-minimal#27](#)

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This was referenced 3 weeks ago

[Bump versions of multiple dependencies to address vulnerabilities](#) [muditsinha01/multi-lang-minimal#28](#)

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[Bump versions of multiple dependencies to address vulnerabilities](#) [muditsinha01/multi-lang-minimal#29](#)

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This was referenced 2 weeks ago

[Bump versions of multiple dependencies to address vulnerabilities](#) [muditsinha01/multi-lang-minimal#30](#)

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vulnerabilities [muditsinha01/multi-lang-minimal#43](#)

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vulnerabilities [muditsinha01/multi-lang-minimal#44](#)



This was referenced 10 hours ago

Bump versions of multiple dependencies to address

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vulnerabilities muditsinha01/multi-lang-minimal#45

▯ **Bump versions of multiple dependencies to address**

vulnerabilities muditsinha01/multi-lang-minimal#46

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